A Note on the Smoluchowski–Kramers Approximation with Constant Friction: A Revised Derivation & Validation*

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April 18, 2026

1 Introduction

Newton's second law of motion is one of the foundational principles in classical mechanics. It states that the acceleration of an object is directly proportional to the net external force acting upon it and inversely proportional to its mass. Mathematically, this relationship is expressed as:

$$\mathbf{F}_{\text{net}} = m\mathbf{a} = m\frac{d^2\mathbf{s}(t)}{dt^2} \tag{1}$$

where $\mathbf{F}_{\text{net}}$ is the total net force acting on the object, m is its mass, $\mathbf{a}$ is the resulting acceleration, and $\mathbf{s}(t)$ describes the position of the object as a function of time. In this work, we explore how the classical second-order dynamics described by Newton's law reduce to a first-order stochastic differential equation as the object's mass goes to zero. This limiting procedure, known as the *Smoluchowski-Kramers approximation.*

We consider the motion of a particle of mass μ in a deterministic force field $b(q)$, where $q \in \mathbb{R}^n$, and subject to both random fluctuations modeled by Gaussian white noise and a friction force proportional to the velocity. Denote the position of the particle q_t^μ. According to Newton's second law, the dynamics of the particle are governed by the second-order stochastic differential equation:

$$\mu\ddot{q}_t^\mu = b(q_t^\mu) + \sigma(q_t^\mu)\dot{W}_t - \alpha\dot{q}_t^\mu, \quad q_0^\mu = q \in \mathbb{R}^n, \quad \dot{q}_0^\mu = p \in \mathbb{R}^n \tag{2}$$

In Eq. (2), $b(q)$ is the deterministic vector field, $\dot{W}_t$ is a Gaussian White Noise process in $\mathbb{R}^n$, and $\sigma(q)$ is an $n \times n$ matrix. The term $\alpha\dot{q}_t^\mu$ describes the friction to the motion that is proportional to the velocity. The functions $b(q)$ and $\sigma(q)$ are supposed to be regular satisfying Assumption 1, so that the solution of Eq. (2) exists and is unique.

*This note provides a detailed proof of the Smoluchowski–Kramers Approximation. This work was done in my Undergraduate study during the Mathematics Seminar course at Fulbright University Vietnam under Prof. Truong-Son Van. All remaining errors in this note are, of course, my own.

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It is usually assumed in Eq. (2) the friction coefficient α is a positive constant. Under this assumption, one can prove that q_t^μ converges in probability as $\mu \downarrow 0$ uniformly on each finite time interval $[0, T]$ to an n-dimensional Itô diffusion process q_t.

$$\dot{q}_t = \frac{1}{\alpha} b(q_t) + \frac{1}{\alpha} \sigma(q_t) \dot{W}_t, \quad q_0 = q_0^\mu = q \in \mathbb{R}^n \tag{3}$$

In particular, consider the second-order Newton equation (2) and its first-order limiting counterpart (3). In Freidlin (2004), it was established that on each finite time interval $[0, T]$, for any $\epsilon > 0$, and any initial conditions $p, q \in \mathbb{R}^n$, one has:

$$\lim_{\mu \downarrow 0} \mathbb{P}\left(\max_{0 \le t \le T} |q_t^\mu - q_t| > \epsilon \right) = 0 \tag{4}$$

The approximation of q_t^μ by q_t for $0 < \mu \ll 1$ in Eq. (4) is called the *Smoluchowski–Kramers* approximation which justifies replacing the second-order (acceleration) equation with a first-order (velocity) equation in physical experiments.

In this note, we provide a complete, step-by-step derivation of Eq. (4). We carefully revisit the original argument, correct an inaccuracy in the main result, establish a revised convergence rate, and validate our theoretical findings through numerical simulations.

The key idea behind this approximation lies in the physical interpretation of the inertial term $\mu \ddot{q}_t^\mu$, which represents the effect of mass times acceleration. When the mass μ is very small, this term becomes negligible compared to the dominant forces—friction and noise. Physically, when mass is negligible, friction instantly adjusts the velocity to match the applied forces and noise, thus acceleration no longer contributes significantly to the dynamics. This eliminates the need to track acceleration separately, allowing one to describe the motion entirely in terms of position and velocity.

2 Result

We begin with the following conjecture to develop intuition about the relationship between the second-order differential equation derived from Newton's second law and their corresponding first-order stochastic limit. Without loss of generality, we assume the constant friction coefficient $\alpha = 1$.

Conjecture 2.1. *Consider the second-order differential equation arising from Newton's second law:*

$$\mu \ddot{q}_t^\mu = b(t, q_t^\mu) + \sigma(t, q_t^\mu)\dot{W}_t - \dot{q}_t^\mu, \quad q_0^\mu = q, \quad \dot{q}_0^\mu = p, \tag{5}$$

and the associated Itô stochastic differential equation:

$$\dot{q}_t = b(t, q_t) + \sigma(t, q_t)\dot{W}_t, \quad q_0 = q. \tag{6}$$

Then, both equations can be respectively expressed that:

$$q_t^\mu = q + \int_0^t b(s, q_s^\mu)\, ds + \int_0^t \sigma(s, q_s^\mu)\, dW_s + \underbrace{\cdots\cdots\cdots\cdots\cdots\cdots}_{\leq D\mu},$$

$$q_t = q + \int_0^t b(s, q_s)\, ds + \int_0^t \sigma(s, q_s)\, dW_s.$$

It is conjectured that the residual terms, $|q_t^\mu - q_t|$, are bounded in magnitude by $D\mu$ for some constant $D > 0$. Therefore, as the mass $\mu \to 0$, the difference between the trajectories q_t^μ and q_t becomes arbitrarily small.

To analyze this convergence, we make the following assumptions on the drift and diffusion coefficients.

Assumption 1. *It is assumed that both $b(t, q)$ and $\sigma(t, q)$ Lipschitz continuous in q with constant K, and bounded uniformly in t and q.*

- *Lipschitz continuity in q with constant $K < \infty$:*

  $$|b(t, q_1) - b(t, q_2)| \leq K|q_1 - q_2| \quad \text{and} \quad |\sigma(t, q_1) - \sigma(t, q_2)| \leq K|q_1 - q_2|$$

  for all $q_1, q_2 \in \mathbb{R}^n$, where $|\cdot|$ denotes the Euclidean norm on $\mathbb{R}^n$ for b and the operator norm on $\mathbb{R}^{n\times n}$ for σ.

- *Bounded uniformly in t and q:*

  $$|b(t, q)| \leq N \quad \text{and} \quad |\sigma(t, q)| \leq N$$

  for all for all $t \in [0, T]$ and $q \in \mathbb{R}^n$, where $N < \infty$ is a constant.

These regularity conditions ensure the existence and uniqueness of strong solutions to both the second-order system (5) and the limiting first-order Itô SDE (6). Moreover, they provide stability and control over the nonlinearities in the dynamics, which are essential for proving convergence results as $\mu \to 0$.

The following lemma is from Freidlin (2004). We will revisit this result with a corrected formulation.

Lemma 2.2 (Freidlin (2004)). *Let q_t^μ and q_t be defined by Eqs. (5) and (6) respectively.*

(i) Assuming $\sigma(t,q)=0$ in both equations, then for any $T>0$

$$\max_{0\le t\le T}|q_t^\mu - q_t| \le e^{KT}\mu\left[\sup_{0\le t\le T, q\in\mathbb{R}^n}|b(t,q)|+|p|\right]$$

(ii) There exists $c_1>0$ defined by T,n and the Lipschitz constant K of the coefficients $b(t,q)$ and $\sigma(t,q)$, such that

$$\max_{0\le t\le T}\mathbb{E}\left[|q_t^\mu - q_t|^2\right] \le c_1\mu\left[|p|^2+\|b\|^2+\|\sigma\sigma^\top\|\right]$$

$\|b\| = \sup_{0\le t\le\infty, q\in\mathbb{R}^n, 1\le i\le n}|b_i(t,q)|$ *and* $\|\sigma\sigma^\top\| = \sup_{0\le t\le\infty, q\in\mathbb{R}^n, 1\le i,j\le n}\sum_{k=1}^n|\sigma_{ik}(t,q)\sigma_{kj}(t,q)|$.

(iii) For any $\epsilon>0$ and $T>0$, there exists a constant $c_2=(c_1,n)$ such that

$$\mathbb{P}\left(\max_{0\le t\le T}|q_t^\mu - q_t|>\epsilon\right) \le \frac{\mu}{\epsilon^2}c_2T\left[|p|^2+\|b\|^2+\|\sigma\sigma^\top\|\right]$$

First, we transform both Eqs. (5) and (6) into their respective integral forms. Denoting $\dot{q}_t^\mu = p_t^\mu$, we transform Eq. (5) into a linear first order ODE

$$\dot{p}_t^\mu + \frac{1}{\mu}p_t^\mu = \frac{1}{\mu}b(t,q_t^\mu) + \frac{1}{\mu}\sigma(t,q_t^\mu)\dot{W}_t, \quad p_0^\mu = p$$

Notice that this is a first-order linear equation that can be solved using Integrating factor method

$$\dot{y} + p(t)y = f(t), \quad y(x_0) = y_0$$

with the corresponding solution as

$$y_t = e^{-\int_{x_0}^t p(s)ds}\left(\int_0^t e^{\int_{x_0}^s p(u)du}f(s)ds + y_0\right)$$

It follows that the solution is

$$p_t^\mu = pe^{\frac{-t}{\mu}} + \frac{1}{\mu}e^{\frac{-t}{\mu}}\int_0^t e^{\frac{s}{\mu}}b(s,q_s^\mu)ds + \frac{1}{\mu}e^{\frac{-t}{\mu}}\int_0^t e^{\frac{s}{\mu}}\sigma(s,q_s^\mu)dW_s$$

Since $p_t^\mu = \dot{q}_t^\mu$. Integrating this equation, we get

$$q_t^\mu = q_0^\mu + \int_0^t p_s^\mu\,ds = q + \int_0^t p_s^\mu\,ds$$

With Integration by parts, we obtain the final Eq. (5) as

$$q_t^\mu = q + p\mu\left(1-e^{-\frac{t}{\mu}}\right) + \left[-e^{-\frac{t}{\mu}}\int_0^t e^{\frac{s}{\mu}}b(s,q_s^\mu)\,ds + \int_0^t b(s,q_s^\mu)\,ds\right] + \left[-e^{-\frac{t}{\mu}}\int_0^t e^{\frac{s}{\mu}}\sigma(s,q_s^\mu)\,dW_s + \int_0^t \sigma(s,q_s^\mu)\,dW_s\right]$$

And the corresponding integral version of Itô diffusion stochastic differential equation in Eq. (6) is

$$q_t = q + \int_0^t b(s,q_s)ds + \int_0^t \sigma(s,q_s)dW_s$$

Now, we are well ready to prove each statement in the Lemma 2.2.

1. *Proof.* Assuming the diffusion part $\sigma(t,q)=0$ in both equations, we get:

$$
\begin{aligned}
|q_t^\mu - q_t| &= \left| p\mu\left(1-e^{-\frac{t}{\mu}}\right) + \int_0^t \left[b(s,q_s^\mu) - b(s,q_s)\right] ds - e^{-\frac{t}{\mu}} \int_0^t e^{\frac{s}{\mu}} b(s,q_s^\mu)\, ds \right| \\
&\leq \left| p\mu\left(1-e^{-\frac{t}{\mu}}\right)\right| + \left| \int_0^t \left[b(s,q_s^\mu) - b(s,q_s)\right] ds \right| + \left| e^{-\frac{t}{\mu}} \int_0^t e^{\frac{s}{\mu}} b(s,q_s^\mu)\, ds \right| \\
&\leq |p|\,\mu\left(1-e^{-\frac{t}{\mu}}\right) + \int_0^t K|q_s^\mu - q_s|\, ds + \left| e^{-\frac{t}{\mu}} \int_0^t e^{\frac{s}{\mu}} \sup_{0\leq t\leq T, q\in\mathbb{R}^n} b(t,q)\, ds \right| \\
&\leq |p|\mu + K\int_0^t |q_s^\mu - q_s| ds + \mu \sup_{0\leq t\leq T, q\in\mathbb{R}^n} |b(t,q)| \\
&= \mu\left(|p| + \sup_{0\leq t\leq T, q\in\mathbb{R}^n} b(t,q)\right) + K\int_0^t |q_s^\mu - q_s| ds
\end{aligned}
\tag{7}
$$

where we have used the uniformly boundedness of $b(t,q)$ to obtain:

$$
\begin{aligned}
\left| e^{-\frac{t}{\mu}} \int_0^t e^{\frac{s}{\mu}} b(s,q_s^\mu)\, ds \right| &\leq \left| e^{-\frac{t}{\mu}} \int_0^t e^{\frac{s}{\mu}} \sup_{0\leq t\leq T, q\in\mathbb{R}^n} b(t,q)\, ds \right| = \left| \mu \sup_{0\leq t\leq T, q\in\mathbb{R}^n} b(t,q)\left(1-e^{-\frac{t}{\mu}}\right) \right| \\
&\leq \mu \sup_{0\leq t\leq T, q\in\mathbb{R}^n} |b(t,q)|
\end{aligned}
\tag{8}
$$

Applying the Grownwall Inequality (see Karatzas and Shreve (1998b) p. 287) to Eq. (10) to Eq. (7) leads to

$$
\begin{aligned}
|q_t^\mu - q_t| &\leq \mu\left(|p| + \sup_{0\leq t\leq T, q\in\mathbb{R}^n} b(t,q)|\right) \exp\left(\int_0^t K ds\right) \\
&\leq e^{KT}\mu\left[|p| + \sup_{0\leq t\leq T, q\in\mathbb{R}^n} |b(t,q)|\right], \quad \forall t\in[0,T]
\end{aligned}
$$

Since this inequality is valid for all $t\in[0,T]$, it follows that

$$
\max_{0\leq t\leq T} |q_t^\mu - q_t| \leq e^{KT}\mu\left[|p| + \sup_{0\leq t\leq T, q\in\mathbb{R}^n} |b(t,q)|\right] \tag{9}
$$

Thus statement (i) is proved.

□

2. *Proof.* Similar to the proof of statement (i), by Cauchy-Swartz Inequality, Lipschitz Continuity, Uniformly Boundedness and Itô Isometry, we can obtain the following bound:

$$
\begin{aligned}
&\mathbb{E}\left[|q_t^\mu - q_t|^2\right] \\
&\leq 5\left|p\mu\left(1-e^{-\frac{t}{\mu}}\right)\right|^2 + 5\mathbb{E}\left[\left|\int_0^t \left[b(s,q_s^\mu)-b(s,q_s)\right]\,ds\right|^2\right] + 5\mathbb{E}\left[\left|\int_0^t \left[\sigma(s,q_s^\mu)-\sigma(s,q_s)\right]\,dW_s\right|^2\right] \\
&\quad + 5\left|e^{-\frac{t}{\mu}}\int_0^t e^{\frac{s}{\mu}} b(s,q_s^\mu)\,ds\right|^2 + 5\mathbb{E}\left[\left|e^{-\frac{t}{\mu}}\int_0^t e^{\frac{s}{\mu}}\sigma(s,q_s^\mu)\,dW_s\right|^2\right] \\
&\leq 5|p\mu|^2 + 5t\mathbb{E}\left[\int_0^t \left|b(s,q_s^\mu)-b(s,q_s)\right|^2\,ds\right] + 5\mathbb{E}\left[\int_0^t \left|\sigma(s,q_s^\mu)-\sigma(s,q_s)\right|^2\,ds\right] \\
&\quad + 5\mu^2\|b\|^2 + 5\mathbb{E}\left[\left|e^{-\frac{t}{\mu}}\right|^2\int_0^t \left|e^{\frac{s}{\mu}}\sigma(s,q_s^\mu)\right|^2\,ds\right] \\
&\leq 5|p\mu|^2 + 5T\mathbb{E}\left[\int_0^t K^2|q_s^\mu - q_s|^2 ds\right] + 5\mathbb{E}\left[\int_0^t K^2|q_s^\mu - q_s|^2 ds\right] + 5\mu^2\|b\|^2 + 5\|\sigma\sigma^\top\|^2\frac{\mu}{2} \\
&= 5\mu\left(|p|^2\mu + \mu\|b\|^2 + \frac{1}{2}\|\sigma\sigma^\top\|^2\right) + (5T+1)K^2\int_0^t \mathbb{E}\left[|q_s^\mu - q_s|^2\right]ds
\end{aligned}
\tag{10}
$$

Again we bound the last term by Itô Isometry and the uniformly boundedness of $\sigma(t,q)$:

$$
\begin{aligned}
\mathbb{E}\left[\left|e^{-\frac{t}{\mu}}\int_0^t e^{\frac{s}{\mu}}\sigma(s,q_s^\mu)\,dW_s\right|^2\right] &= \mathbb{E}\left[\left|e^{-\frac{t}{\mu}}\right|^2\int_0^t \left|e^{\frac{s}{\mu}}\sigma(s,q_s^\mu)\right|^2\,ds\right] \leq \mathbb{E}\left[e^{-\frac{2t}{\mu}}\int_0^t e^{\frac{2s}{\mu}}\|\sigma\sigma^\top\|^2\,ds\right] \\
&= \|\sigma\sigma^\top\|^2\mathbb{E}\left[\frac{\mu}{2}\left(1-e^{-\frac{2t}{\mu}}\right)\right] \\
&\leq \|\sigma\sigma^\top\|^2\frac{\mu}{2}
\end{aligned}
\tag{11}
$$

Applying the Grownwall Inequality, we have

$$
\begin{aligned}
\mathbb{E}\left[|q_t^\mu - q_t|^2\right] &\leq 5\mu\left(|p|^2\mu + \mu\|b\|^2 + \frac{1}{2}\|\sigma\sigma^\top\|^2\right)\exp\left(\int_0^t (5T+1)K^2\right)ds \\
&\leq \underbrace{e^{(5T+1)TK^2}5}_{:=c_1}\mu\left(|p|^2\mu + \mu\|b\|^2 + \frac{1}{2}\|\sigma\sigma^\top\|^2\right) \\
&= c_1\mu\left(|p|^2\mu + \mu\|b\|^2 + \frac{1}{2}\|\sigma\sigma^\top\|^2\right), \quad \forall t\in[0,T]
\end{aligned}
$$

As a result, we obtain the statement (ii):

$$
\max_{0\leq t\leq T}\mathbb{E}\left[|q_t^\mu - q_t|^2\right] \leq c_1\mu\left(|p|^2\mu + \mu\|b\|^2 + \frac{1}{2}\|\sigma\sigma^\top\|^2\right) = O(\mu) \tag{12}
$$

□

3. *Proof.* Consider the following difference

$$
\begin{aligned}
q_t^\mu - q_t = p\mu\left(1 - e^{-\frac{t}{\mu}}\right) + \int_0^t \left[b(s, q_s^\mu) - b(s, q_s)\right] ds + \int_0^t \left[\sigma(s, q_s^\mu) - \sigma(s, q_s)\right] dW_s \\
- e^{-\frac{t}{\mu}} \int_0^t e^{\frac{s}{\mu}} b(s, q_s^\mu)\, ds - e^{-\frac{t}{\mu}} \int_0^t e^{\frac{s}{\mu}} \sigma(s, q_s^\mu)\, dW_s
\end{aligned} \tag{13}
$$

We decompose the difference into four principal components:

- Drift term: $X_t = \int_0^t \left[b(s, q_s^\mu) - b(s, q_s)\right] ds$
- Martingale term: $I_t = \int_0^t \left[\sigma(s, q_s^\mu) - \sigma(s, q_s)\right] dW_s$ where I_t is a martingale, so $|I_t|^2$ is a submartingale.
- Weighted/Scaled Martingale:

$$
M_t = -e^{-\frac{t}{\mu}} \int_0^t e^{\frac{s}{\mu}} \sigma(s, q_s^\mu)\, dW_s = -e^{-\frac{t}{\mu}} N_t
$$

  where N_t is a martingale, and is scaled by $-e^{-\frac{t}{\mu}}$.
- Remaining terms: $O(\mu)$

This yields the compact decomposition of Eq. (13) :

$$
q_t^\mu - q_t = X_t + I_t + M_t + O(\mu) \tag{14}
$$

It follows that

$$
\mathbb{E}\left[\max_{0\le t\le T} |q_t^\mu - q_t|^2\right] \le 4\mathbb{E}\left[\max_{0\le t\le T} |X_t|^2\right] + 4\mathbb{E}\left[\max_{0\le t\le T} |I_t|^2\right] + 4\mathbb{E}\left[\max_{0\le t\le T} |M_t|^2\right] + O(\mu^2) \tag{15}
$$

Let us consider the terms separately.

$$
\begin{aligned}
\mathbb{E}\left[\max_{0\le t\le T} |X_t|^2\right] &= \mathbb{E}\left[\max_{0\le t\le T} \left|\int_0^t \left[b(s, q_s^\mu) - b(s, q_s)\right] ds\right|^2\right] \le \mathbb{E}\left[\max_{0\le t\le T} \left\{t \int_0^t |b(s, q_s^\mu) - b(s, q_s)|^2\, ds\right\}\right] \\
&\le \mathbb{E}\left[\max_{0\le t\le T} \left\{tK^2 \int_0^t |q_s^\mu - q_s|^2\, ds\right\}\right]
\end{aligned}
$$

Since the integral non-decreasing in t, so the bound for the maximum satisfies:

$$
\begin{aligned}
\mathbb{E}\left[\max_{0\le t\le T} |X_t|^2\right] &\le \mathbb{E}\left[TK^2 \int_0^T |q_s^\mu - q_s|^2\, ds\right] = TK^2 \int_0^T \mathbb{E}\left[|q_s^\mu - q_s|^2\right] ds \\
&\le c_1\mu\left(|p|^2\mu + \mu\|b\|^2 + \frac{1}{2}\|\sigma\sigma^\top\|^2\right) T^2K^2 = O(\mu)
\end{aligned}
$$

For the second term, by the Doob-Maximal inequality (see (Karatzas and Shreve, 1998a) p.14)

$$\begin{aligned}
\mathbb{E}\left[\max_{0\le t\le T}|I_t|^2\right] &\le 4\mathbb{E}[|I_T|^2] = 4\mathbb{E}\left[\left|\int_0^T \left[\sigma(s,q_s^\mu)-\sigma(s,q_s)\right]\,dW_s\right|^2\right] \le 4\mathbb{E}\left[\int_0^T K^2|q_s^\mu - q_s|^2 ds\right] \\
&= 4K^2\int_0^T \mathbb{E}\left[|q_s^\mu - q_s|^2\right] ds \le 4K^2\int_0^T c_1\mu\left(|p|^2\mu + \mu\|b\|^2 + \frac{1}{2}\|\sigma\sigma^\top\|^2\right) ds \\
&= c_1\mu\left(|p|^2\mu + \mu\|b\|^2 + \frac{1}{2}\|\sigma\sigma^\top\|^2\right) 4K^2T = O(\mu)
\end{aligned}$$

To upper bound the third term, we define the rescaled time $\tau := t/\mu$, the total horizon $S := T/\mu$, and the rescaled Brownian motion $\tilde{W}_\tau = \mu^{-1/2}W_{\mu\tau}$. Setting $r = s/\mu$ in the integral, we express $M_{\mu\tau}$ as

$$M_{\mu\tau} = -\int_0^{\mu\tau} e^{-(\mu\tau - s)/\mu}\sigma(s,q_s^\mu)\,dW_s = -\mu^{1/2}\int_0^\tau e^{-(\tau-r)}\sigma(\mu r, q_{\mu r}^\mu)\,d\tilde{W}_r.$$

We define the rescaled process $\tilde{M}_\tau$ by:

$$\tilde{M}_\tau := \mu^{-1/2}M_{\mu\tau} = -\int_0^\tau e^{-(\tau-r)}\sigma(\mu r, q_{\mu r}^\mu)\,d\tilde{W}_r,$$

such that $M_{\mu\tau} = \mu^{1/2}\tilde{M}_\tau$. Therefore, the expected maximum satisfies:

$$\mathbb{E}\left[\max_{0\le t\le T}|M_t|^2\right] = \mu\cdot\mathbb{E}\left[\max_{0\le\tau\le S}|\tilde{M}_\tau|^2\right]$$

For $\tau > \tau'$, the difference of the process is

$$\begin{aligned}
\tilde{M}_\tau - \tilde{M}_{\tau'} &= -\int_0^\tau e^{-(\tau-r)}\sigma(\mu r, q_{\mu r}^\mu)\,d\tilde{W}_r + \int_0^{\tau'} e^{-(\tau'-r)}\sigma(\mu r, q_{\mu r}^\mu)\,d\tilde{W}_r \\
&= -\left[\int_0^{\tau'} e^{-(\tau-r)}\sigma(\mu r, q_{\mu r}^\mu)\,d\tilde{W}_r + \int_{\tau'}^\tau e^{-(\tau-r)}\sigma(\mu r, q_{\mu r}^\mu)\,d\tilde{W}_r\right] + \int_0^{\tau'} e^{-(\tau'-r)}\sigma(\mu r, q_{\mu r}^\mu)\,d\tilde{W}_r \\
&= -\int_0^{\tau'}\left[e^{-(\tau-r)} - e^{-(\tau'-r)}\right]\sigma(\mu r, q_{\mu r}^\mu)\,d\tilde{W}_r - \int_{\tau'}^\tau e^{-(\tau-r)}\sigma(\mu r, q_{\mu r}^\mu)\,d\tilde{W}_r \\
&= -\int_0^{\tau'}\left[e^{-(\tau'-r)}\cdot e^{-(\tau-\tau')} - e^{-(\tau'-r)}\right]\sigma(\mu r, q_{\mu r}^\mu)\,d\tilde{W}_r - \int_{\tau'}^\tau e^{-(\tau-r)}\sigma(\mu r, q_{\mu r}^\mu)\,d\tilde{W}_r \\
&= \int_0^{\tau'}\left[e^{-(\tau'-r)}\left(1 - e^{-(\tau-\tau')}\right)\right]\sigma(\mu r, q_{\mu r}^\mu)\,d\tilde{W}_r - \int_{\tau'}^\tau e^{-(\tau-r)}\sigma(\mu r, q_{\mu r}^\mu)\,d\tilde{W}_r
\end{aligned}$$

By the Itô isometry, for each term and using the inequality $1 - e^{-y} \leq \min(y, 1)$ for all $y \geq 0$, we get:

$$\begin{aligned}
\mathbb{E}\left[\left|\int_0^{\tau'} \left[e^{-(\tau'-r)}\left(1 - e^{-(\tau-\tau')}\right)\right] \sigma(\mu r, q^{\mu}_{\mu r})\, d\tilde{W}_r\right|^2\right] &= \mathbb{E}\left[\int_0^{\tau'} \left|e^{-(\tau'-r)}\left(1 - e^{-(\tau-\tau')}\right)\right|^2 \left|\sigma(\mu r, q^{\mu}_{\mu r})\right|^2 dr\right] \\
&\leq \int_0^{\tau'} e^{-2(\tau'-r)} \left[1 - e^{-(\tau-\tau')}\right]^2 \|\sigma\sigma^\top\|^2\, dr \\
&\leq \|\sigma\sigma^\top\|^2 \left[1 - e^{-(\tau-\tau')}\right]^2 \int_0^{\tau'} e^{-2(\tau'-r)}\, dr \\
&\leq \frac{\|\sigma\sigma^\top\|^2}{2} \left[1 - e^{-(\tau-\tau')}\right]^2 (1 - e^{-2\tau'}) \\
&\leq \frac{\|\sigma\sigma^\top\|^2}{2} \min\left((\tau - \tau')^2, 1\right)
\end{aligned}$$

And

$$\begin{aligned}
\mathbb{E}\left[\left|\int_{\tau'}^{\tau} e^{-(\tau-r)} \sigma(\mu r, q^{\mu}_{\mu r})\, d\tilde{W}_r\right|^2\right] &= \mathbb{E}\left[\int_{\tau'}^{\tau} e^{-2(\tau-r)} \left|\sigma(\mu r, q^{\mu}_{\mu r})\right|^2 dr\right] \\
&\leq \|\sigma\sigma^\top\|^2 \int_{\tau'}^{\tau} e^{-2(\tau-r)}\, dr = \frac{\|\sigma\sigma^\top\|^2}{2}\left(1 - e^{-2(\tau-\tau')}\right) \\
&\leq \frac{\|\sigma\sigma^\top\|^2}{2} \min\left(2(\tau - \tau'), 1\right)
\end{aligned}$$

In the end, we obtain the following

$$\begin{aligned}
\mathbb{E}\left[|\tilde{M}_\tau - \tilde{M}_{\tau'}|^2\right] &\leq \frac{\|\sigma\sigma^\top\|^2}{2} \min\left((\tau - \tau')^2, 1\right) + \frac{\|\sigma\sigma^\top\|^2}{2} \min\left(2(\tau - \tau'), 1\right) \\
&\leq \frac{\|\sigma\sigma^\top\|^2}{2} 3 \min(\tau - \tau', 1) \\
&:= B \min(\tau - \tau', 1)
\end{aligned}$$

which we can define a pseudometric for all $\tau, \tau' \geq 0$ as:

$$d_{\tilde{M}}(\tau, \tau') := \sqrt{\mathbb{E}\left[|\tilde{M}_\tau - \tilde{M}_{\tau'}|^2\right]} \leq \sqrt{B \min(|\tau - \tau'|, 1)} =: \rho(\tau, \tau')$$

In addition, we can express the difference as the following

$$\begin{aligned}
&\tilde{M}_\tau - \tilde{M}_{\tau'} = \int_0^{\tau} g(r)\, d\tilde{W}_r, \text{ where:} \\
&g(r) = -\left[e^{-(\tau-r)} - e^{-(\tau'-r)}\right] \sigma(\mu r, q^{\mu}_{\mu r}) \mathbf{1}_{\{r \leq \tau'\}} - e^{-(\tau-r)} \sigma(\mu r, q^{\mu}_{\mu r}) \mathbf{1}_{\{\tau' < r \leq \tau\}}
\end{aligned}$$

Following the same calculation, it holds almost surely that:

$$\int_0^{\tau} |g(r)|^2\, dr = \int_0^{\tau'} \left|e^{-(\tau-r)} - e^{-(\tau'-r)}\right|^2 \left|\sigma(\mu r, q^{\mu}_{\mu r})\right|^2 dr + \int_{\tau'}^{\tau} e^{-2(\tau-r)} \left|\sigma(\mu r, q^{\mu}_{\mu r})\right|^2 dr \leq B \min(|\tau - \tau'|, 1)$$

Let $\lambda \in \mathbb{R}$, according to Novikov's condition (e.g see Pham (2009) p. 21)

$$\mathbb{E}\left[e^{\frac{1}{2}\int_0^\tau |\lambda g(r)|^2\,dr}\right] = \mathbb{E}\left[e^{\frac{\lambda^2}{2}\int_0^\tau |g(r)|^2\,dr}\right] \le \mathbb{E}\left[e^{\frac{\lambda^2}{2}B\min(|\tau-\tau'|,1)}\right] = e^{\frac{\lambda^2}{2}B\min(|\tau-\tau'|,1)} < \infty$$

It follows that the process

$$\mathcal{E}_t := \exp\left(\int_0^t \lambda g(r)\,d\tilde{W}_r - \frac{1}{2}\int_0^t \left|\lambda g(r)\right|^2\,dr\right), \quad 0 \le t \le \tau$$

is a martingale. Therefore $\mathbb{E}[\mathcal{E}_\tau] = \mathbb{E}[\mathcal{E}_0] = \mathbb{E}[\exp(0)] = 1$. As a result, we obtain

$$\begin{aligned}
\mathbb{E}\left[e^{\lambda(\tilde{M}_\tau - \tilde{M}_{\tau'})}\right] &= \mathbb{E}\left[e^{\lambda\int_0^\tau g(r)\,d\tilde{W}_r}\right] = \mathbb{E}\left[e^{\frac{\lambda^2}{2}\int_0^\tau |g(r)|^2\,dr}\cdot \underbrace{e^{\lambda\int_0^\tau g(r)\,d\tilde{W}_r - \frac{\lambda^2}{2}\int_0^\tau |g(r)|^2\,dr}}_{=\mathcal{E}_\tau}\right] \\
&\le e^{\frac{\lambda^2}{2}B\min(|\tau-\tau'|,1)}\cdot \mathbb{E}[\mathcal{E}_\tau] = e^{\frac{\lambda^2}{2}B\min(|\tau-\tau'|,1)}\cdot 1 \\
&= \exp\left(\frac{\lambda^2\rho(\tau,\tau')^2}{2}\right)
\end{aligned} \tag{16}$$

For any $u > 0$, we apply Chernoff bound (e.g see Pishro-Nik (2014)). For the upper tail, we have:

$$\mathbb{P}(\tilde{M}_\tau - \tilde{M}_{\tau'} \ge u) \le \inf_{\lambda>0}\left(e^{-\lambda u}\mathbb{E}\left[\exp\left(\lambda(\tilde{M}_\tau - \tilde{M}_{\tau'})\right)\right]\right) \le \inf_{\lambda>0}\exp\left(\frac{\lambda^2\rho(\tau,\tau')^2}{2} - \lambda u\right).$$

To find the tightest bound, we minimize the term inside the exponent. Taking the derivative with respect to λ and setting it to zero, gives $\lambda = u/\rho(\tau,\tau')^2$. Substituting this optimal value of λ back into the exponent yields the one-sided tail bound

$$\mathbb{P}(\tilde{M}_\tau - \tilde{M}_{\tau'} \ge u) \le \exp\left(-\frac{u^2}{2\rho(\tau,\tau')^2}\right).$$

Since we are interested in the two-sided probability $\mathbb{P}(|\tilde{M}_\tau - \tilde{M}_{\tau'}| \ge u)$, we apply the same bound to the lower tail $\mathbb{P}(\tilde{M}_\tau - \tilde{M}_{\tau'} \le -u)$ (noting the symmetry of the bound with respect to the sign):

$$\mathbb{P}\left(|\tilde{M}_\tau - \tilde{M}_{\tau'}| \ge u\right) \le 2\exp\left(-\frac{u^2}{2\rho(\tau,\tau')^2}\right) =: 2\exp\left(-\frac{u^2}{cM^2}\right) \tag{17}$$

By Remark 2.2 in Viens and Vizcarra (2007), with constant $c = 1/3$, Eq. (17) implies the following equation with $M = \sqrt{6}\rho(\tau,\tau')$

$$\mathbb{E}\left[\exp\left(\left(\frac{\tilde{M}_\tau - \tilde{M}_{\tau'}}{\sqrt{6}\rho(\tau,\tau')}\right)^2\right)\right] = \mathbb{E}\left[\exp\left(\left(\frac{\tilde{M}_\tau - \tilde{M}_{\tau'}}{M}\right)^2\right)\right] \le 2$$

Define a new pseudometric $\delta(\tau,\tau') := \sqrt{6}\rho(\tau,\tau') = \sqrt{6B\min(|\tau-\tau'|,1)} =: \sqrt{C\min(|\tau-\tau'|,1)}$. With reverse triangle inequality, in the end we obtain that

$$\mathbb{E}\left[\exp\left(\left(\frac{\left||\tilde{M}_\tau| - |\tilde{M}_{\tau'}|\right|}{\delta(\tau,\tau')}\right)^2\right)\right] \leq \mathbb{E}\left[\exp\left(\left(\frac{|\tilde{M}_\tau - \tilde{M}_{\tau'}|}{\delta(\tau,\tau')}\right)^2\right)\right] \leq 2 \tag{18}$$

This shows that the process $(|\tilde{M}_\tau|)_{0\leq\tau\leq S}$ satisfies the sub-Gaussian chaos condition (see Definitions 2.1 and 2.3 in Viens and Vizcarra (2007)), for $n=1$ with respect to the new pseudometric $\delta(\tau,\tau')$. Importantly, it satisfies the first condition in the main result of Viens and Vizcarra (2007) in page 3.

Let $J=[0,S]$. The uniform bound for the diameter of J under the metric δ is

$$\operatorname{diam}(J) := \sup_{\tau,\tau'\in[0,S]} \delta(\tau,\tau') = \sup_{\tau,\tau'\in[0,S]} \sqrt{C\cdot\min(|\tau-\tau'|,1)} \leq \sqrt{C}$$

Let $N(\epsilon,J,\delta)$ denote the minimum number of δ-balls of radius ϵ needed to cover J. We distinguish two cases:

Case 1: $\epsilon \geq \sqrt{C}$. Since $\delta(\tau,\tau') \leq \operatorname{diam}(J) \leq \sqrt{C} \leq \epsilon$ for all $\tau,\tau' \in J$, a single ball covers the entire interval. Hence,

$$N(\epsilon,J,\delta) = 1, \quad \text{for } \epsilon \geq \sqrt{C}.$$

Case 2: $0<\epsilon<\sqrt{C}$. The condition $\delta(\tau,\tau')<\epsilon$ requires $\sqrt{C\cdot\min(|\tau-\tau'|,1)}<\epsilon$, implying that

$$\min(|\tau-\tau'|,1) < \frac{\epsilon^2}{C} < 1$$

which forces $\min(|\tau-\tau'|,1) = |\tau-\tau'|$. Thus, $|\tau-\tau'| < \epsilon^2/C$. Covering $[0,S]$ with intervals of length ϵ^2/C requires at most

$$N(\epsilon,J,\delta) \leq \left\lceil \frac{S}{\epsilon^2/C} \right\rceil \leq \frac{CS}{\epsilon^2} + 1 \leq \frac{2CS}{\epsilon^2},$$

where the last inequality holds provided $\epsilon^2 \leq CS$ (which is satisfied for small ϵ or $S\geq 1$, i.e $T \geq \mu$).

According to Viens and Vizcarra (2007)' main results in page 3, Eq. (18) implies there exists a universal constant $C_1>0$ such that

$$\begin{aligned}
\mathbb{E}\left[\max_{0\leq\tau\leq S}|\tilde{M}_\tau|\right] &\leq C_1\int_0^\infty \sqrt{\log N(\epsilon,J,\delta)}\,d\epsilon \\
&= C_1\int_0^{\sqrt{C}} \sqrt{\log N(\epsilon,J,\delta)}\,d\epsilon + C_1\int_{\sqrt{C}}^\infty \sqrt{\log N(\epsilon,J,\delta)}\,d\epsilon \\
&= C_1\int_0^{\sqrt{C}} \sqrt{\log N(\epsilon,J,\delta)}\,d\epsilon = C_1\int_0^{\sqrt{C}} \sqrt{\log\left(\frac{2CS}{\epsilon^2}\right)}\,d\epsilon
\end{aligned}$$

Performing change of variables $\epsilon = \sqrt{C}\,u$, $d\epsilon = \sqrt{C}\,du$

$$\begin{aligned}
\mathbb{E}\left[\max_{0\le\tau\le S}|\tilde{M}_\tau|\right] &\le C_1\sqrt{C}\int_0^1\sqrt{\log\left(\frac{2S}{u^2}\right)}\,du = C_1\sqrt{C}\int_0^1\sqrt{\log(2S)+\log\left(\frac{1}{u^2}\right)}\,du. \\
&\le C_1\sqrt{C}\left(\sqrt{\log(2S)}+\int_0^1\sqrt{\log\left(\frac{1}{u^2}\right)}\,du\right) \\
&= C_1\sqrt{C}\left(\sqrt{\log(2S)}+\sqrt{\frac{\pi}{2}}\right)
\end{aligned}$$

To bound the second moment, we observe that

$$\mathbb{E}\left[\max_{0\le\tau\le S}|\tilde{M}_\tau|^2\right] = \mathbb{E}\left[\max_{0\le\tau\le S}|\tilde{M}_\tau|\right]^2 + \mathbb{V}\left(\max_{0\le\tau\le S}|\tilde{M}_\tau|\right)$$

By Eq. (18), it follows that we get the following concentration inequality with $n = 1$ (e.g see the main result in Viens and Vizcarra (2007)' page 3), and using that $\mathrm{diam}(J) \le \sqrt{C}$:

$$\mathbb{P}\left(\left|\max_{0\le\tau\le S}|\tilde{M}_\tau| - \mathbb{E}\left[\max_{0\le\tau\le S}|\tilde{M}_\tau|\right]\right| > u\right) \le 2\exp\left(-\frac{1}{2}\left(\frac{u}{C_1\mathrm{diam}(J)}\right)^2\right) \le 2\exp\left(-\frac{u^2}{2C_1^2C}\right).$$

Using the integral representation of variance, we have:

$$\begin{aligned}
\mathbb{V}\left(\max_{0\le\tau\le S}|\tilde{M}_\tau|\right) &= \mathbb{E}\left[\left(\max_{0\le\tau\le S}|\tilde{M}_\tau| - \mathbb{E}\left[\max_{0\le\tau\le S}|\tilde{M}_\tau|\right]\right)^2\right] \\
&= \int_0^\infty 2u\cdot\mathbb{P}\left(\left|\max_{0\le\tau\le S}|\tilde{M}_\tau| - \mathbb{E}\left[\max_{0\le\tau\le S}|\tilde{M}_\tau|\right]\right| > u\right)du. \\
&\le \int_0^\infty 2u\cdot 2\exp\left(-\frac{u^2}{2C_1^2C}\right)du \\
&= 4\int_0^\infty u\exp\left(-\frac{u^2}{2C_1^2C}\right)du.
\end{aligned}$$

Making the substitution $v = u^2$, so that $dv = 2u\,du$, we express:

$$\mathbb{V}\left(\max_{0\le\tau\le S}|\tilde{M}_\tau|\right) \le 4\int_0^\infty\frac{1}{2}\exp\left(-\frac{v}{2C_1^2C}\right)dv = 4C_1^2C.$$

Therefore

$$\mathbb{E}\left[\max_{0\le\tau\le S}|\tilde{M}_\tau|^2\right] = \mathbb{E}\left[\max_{0\le\tau\le S}|\tilde{M}_\tau|\right]^2 + \mathbb{V}\left(\max_{0\le\tau\le S}|\tilde{M}_\tau|\right) \le C_1^2C\left(\sqrt{\log(2S)}+\sqrt{\frac{\pi}{2}}\right)^2 + 4C_1^2C.$$

which yields

$$\mathbb{E}\left[\max_{0\le t\le T}|M_t|^2\right] = \mu\cdot\mathbb{E}\left[\max_{0\le \tau\le S}|\tilde{M}_\tau|^2\right] \le \mu\cdot C_1^2 C\left(\sqrt{\log\left(2\cdot\frac{T}{\mu}\right)}+\sqrt{\frac{\pi}{2}}\right)^2 + \mu\cdot 4C_1^2 C.$$
$$= O(\mu\log(1/\mu))$$

Combining all terms in Eq. (15), we get the total convergence rate as

$$\begin{aligned}\mathbb{E}\left[\max_{0\le t\le T}|q_t^\mu - q_t|^2\right] &\le 4\mathbb{E}\left[\max_{0\le t\le T}|X_t|^2\right] + 4\mathbb{E}\left[\max_{0\le t\le T}|I_t|^2\right] + 4\mathbb{E}\left[\max_{0\le t\le T}|M_t|^2\right] + O(\mu^2)\\ &= O\left(\mu+\mu\log(1/\mu)+\mu^2\right)\\ &= O\left(\mu\log(1/\mu)\right)\end{aligned}$$

Applying Chebyshev's inequality yields, for all $\epsilon > 0$:

$$\mathbb{P}\left(\max_{0\le t\le T}|q_t^\mu - q_t| > \epsilon\right) \le \frac{O\left(\mu\log(1/\mu)\right)}{\epsilon^2}$$

which decays to 0 as $\mu \to 0$. □

Remark. *We conclude the note with the following observations regarding Lemma 2.2:*

(a) In the deterministic case where $\sigma(t,q)=0$, the diffusion terms vanish. From the proof of statement (i), we observe that the convergence is in fact uniform and occurs at a linear rate:

$$\max_{0\le t\le T} |q_t^\mu - q_t| = O(\mu).$$

(b) From statement (ii), applying Jensen's inequality yields:

$$\max_{0\le t\le T} \left(\mathbb{E}\left[|q_t^\mu - q_t|\right]\right)^2 \le \max_{0\le t\le T} \mathbb{E}\left[|q_t^\mu - q_t|^2\right] = O(\mu),$$

which implies:

$$\max_{0\le t\le T} \mathbb{E}\left[|q_t^\mu - q_t|\right] = O(\sqrt{\mu}).$$

(c) From statement (iii), we obtain

$$\mathbb{P}\left(\max_{0\le t\le T} |q_t^\mu - q_t| > \epsilon\right) \le \frac{\mathbb{E}\left[\max_{0\le t\le T} |q_t^\mu - q_t|^2\right]}{\epsilon^2} = \frac{O\left(\mu\log(1/\mu)\right)}{\epsilon^2}$$

This implies that the two processes converge pathwise in probability when the mass $\mu \to 0$.

Furthermore, by Jensen's inequality again, we obtain the L^1 convergence rate

$$\mathbb{E}\left[\max_{0\le t\le T} |q_t^\mu - q_t|\right] = O\left(\sqrt{\mu\log(1/\mu)}\right)$$

Comparing the three cases, we obtain a hierarchy of convergence rates as $\mu \to 0$:

$$O(\mu) \ll O\Big(\mu\log\tfrac{1}{\mu}\Big) \ll O(\sqrt{\mu}) \ll O\left(\sqrt{\mu\log\tfrac{1}{\mu}}\right).$$

These correspond respectively to:

i. Deterministic/pathwise uniform rate: $O(\mu)$ (noise-free limit),

ii. Pathwise L^2 rate: $O(\mu\log(1/\mu))$ (bound on $\mathbb{E}[\max_t |q_t^\mu - q_t|^2]$),

iii. Pointwise L^1 rate: $O(\sqrt{\mu})$ (bound on $\max_t \mathbb{E}[|q_t^\mu - q_t|]$),

iv. Pathwise L^1 rate: $O(\sqrt{\mu\log(1/\mu)})$ (bound on $\mathbb{E}[\max_t |q_t^\mu - q_t|]$).

The probability bound $\mathbb{P}(\max_t |q_t^\mu - q_t| > \epsilon) \le O(\mu\log(1/\mu))/\epsilon^2$ follows directly from (b) via Chebyshev's inequality and shares the same asymptotic order as the mean-square rate.

In the stochastic setting, the noise introduces fluctuations that persist even for small μ, slowing down the convergence compared to the deterministic case. Imposing stronger pathwise uniform control while accounting for stochastic noise inherently degrades convergence, reducing the deterministic linear rate $O(\mu)$ to $O\Big(\sqrt{\mu\log(1/\mu)}\Big)$ under the most requirement L^1-pathwise metric.

2.1 Numerical Verification of Convergence Rates

To complement the theoretical analysis, we perform Monte Carlo simulations to empirically validate the convergence rates stated in Remark 3 using the Euler–Maruyama scheme for the coupled system (5)–(6).[1]

We estimate the four error metrics, for $e_t = q_t^\mu - q_t$:

$$\text{(i)} \ \max_t |e_t|, \quad \text{(ii)} \ \max_t \mathbb{E}[|e_t|], \quad \text{(iii)} \ \mathbb{E}[\max_t |e_t|^2], \quad \text{(iv)} \ \mathbb{E}[\max_t |e_t|],$$

and fit power-law models error $\approx C\mu^\alpha$ via log-log linear regression. The empirical exponents $\hat{\alpha}$ is reported in Figure 1.

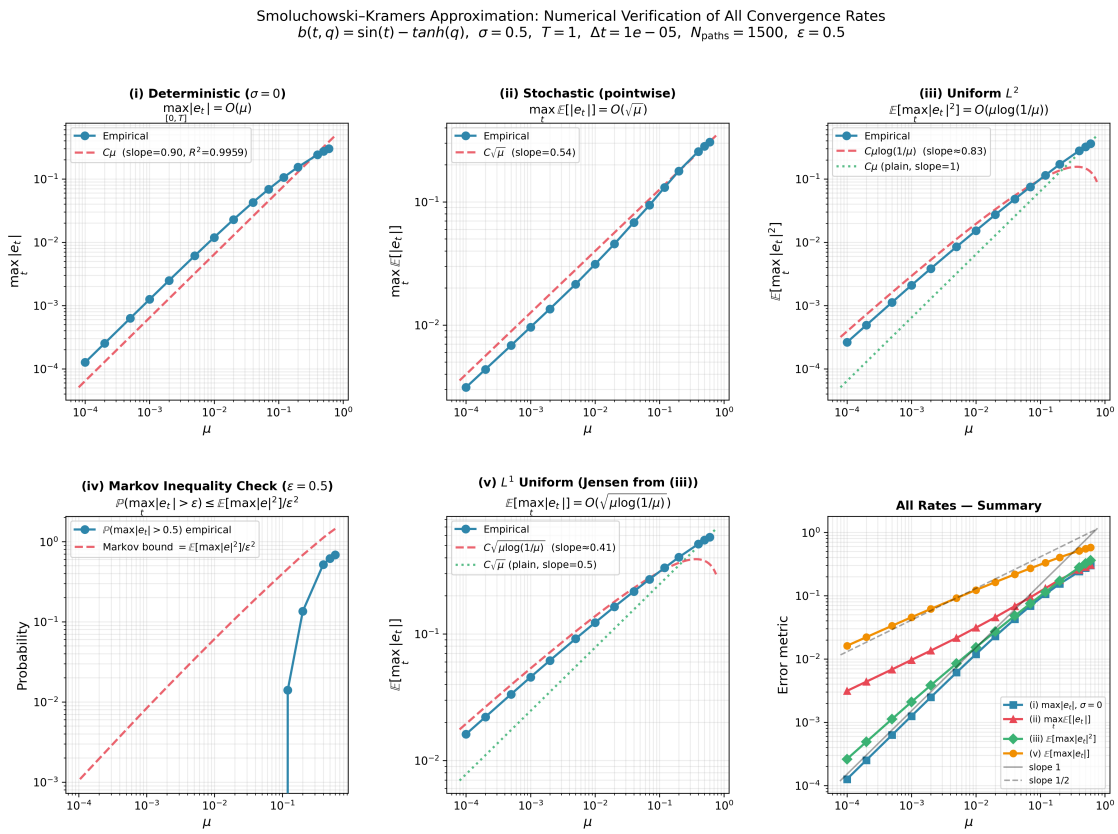

Figure 1: Log–log plots display empirical error metrics with dashed reference lines indicating the theoretical rates $O(\mu)$, $O(\sqrt{\mu})$, $O\big(\mu \log(1/\mu)\big)$, and $O\big(\sqrt{\mu \log(1/\mu)}\big)$. The bottom-right panel illustrates the asymptotic hierarchy: deterministic convergence strictly dominates stochastic pathwise bounds, confirming that noise and uniform-in-time control jointly degrade the rate.

All fitted slopes are very closed to their theoretical counterparts, providing strong numerical evidence that the theoretical asymptotic rates are correct.

[1] The code is available at https://github.com/vuminh-hoangd/Smoluchowski-Kramers-Approximation

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